



Abdou Moumouni University

International Master Program in Energy and Green Hydrogen
(IMP-EGH)



Model Answers

Power-to-X (PtX)

Part 1: Foundations Part 2: Pathways Case-study projects

Worked solutions to the three 1-hour subjects (A, B, C)
with justifications, calculations and the Kassø e-Methanol case study

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Answer Key at a Glance

Power-to-X (PtX)

How to use this document

- The key below lists the expected answers to Parts A and B of each subject (T = True, F = False).
- The following sections give, for every question, the answer and a short justification taken from the course, followed by the completed tables (Part C), the balanced reactions with conditions (Part D) and a full answer to the case study (Part E) for Project 4, Kassø e-Methanol.
- Values written as ranges are typical values from the course and the literature; figures on the Kassø plant come from the developer's public communications (see References).

Part A. Multiple-choice questions

Subject	A1	A2	A3	A4	A5	A6	A7	A8
A	B	D	A	D	B	C	B	D
B	B	A	B	D	B	B	A	B
C	D	C	B	A	D	B	A	A

Part B. True or False

Subject	B1	B2	B3	B4	B5	B6	B7	B8
A	T	T	F	F	F	T	F	T
B	T	F	F	T	T	F	T	F
C	T	F	T	F	T	T	F	T

Key figures used in the answers

- Molar masses: CH₃OH 32.04, H₂ 2.016, CO₂ 44.01, H₂O 18.015 g/mol.
- Stoichiometry of methanol from CO₂: per tonne of CH₃OH, 0.189 t H₂, 1.374 t CO₂ and 0.562 t H₂O by-product.
- Theoretical water need of electrolysis: $18.015/2.016 = 8.94$ kg H₂O per kg H₂.
- Kassø plant: 42,000 t/yr e-methanol, three PEM electrolyzers of about 52 MW in total (Siemens Energy), 304 MW solar park supplying about half of the electricity, biogenic CO₂, RFNBO-certified (2025).

Model Answers | Subject A

Power-to-X (PtX)

Part A. Multiple-choice questions

(8 × 0.5 = 4 pts)

A1. In “Power-to-X”, the letter X designates:

Answer B, the final product (carrier, fuel, chemical or heat).

Justification. “Power” is the renewable electricity input; “X” is the output: hydrogen or a hydrogen-derived carrier (ammonia, methanol, synthetic methane, liquid fuels), a chemical, or heat.

A2. In a PEM electrolyser, the reaction at the anode is:

Answer D, $\text{H}_2\text{O} \longrightarrow \frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^-$.

Justification. The PEM electrolyte is acidic and conducts protons. Water is oxidised at the anode (oxygen evolution); the protons cross the membrane and are reduced at the cathode ($2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$). Option A is the SOEC anode, option C the alkaline cathode, option B a cathode reaction.

A3. The electrolyte of a solid oxide electrolyser (SOEC) is typically:

Answer A, yttria-stabilised zirconia (YSZ).

Justification. YSZ is a ceramic that conducts O^{2-} ions at 600–1000 °C. Nafion is the PEM membrane, 25–30 % KOH the alkaline electrolyte, and molten carbonate the MCFC electrolyte.

A4. Catalyst and pressure of green methanol synthesis:

Answer D, Cu/ZnO-based catalyst, 50–100 bar.

Justification. Methanol synthesis uses Cu/ZnO/Al₂O₃ at about 200–300 °C and 50–100 bar. An iron catalyst at 150–300 bar describes Haber–Bosch; Co or Fe at low pressure describes Fischer–Tropsch.

A5. Reaction that prepares syngas for Fischer–Tropsch from green H₂ and captured CO₂:

Answer B, $\text{H}_2 + \text{CO}_2 \longrightarrow \text{CO} + \text{H}_2\text{O}$, the reverse water-gas shift (RWGS).

Justification. FT synthesis needs CO, not CO₂. The endothermic RWGS ($\Delta H \approx +41$ kJ/mol) converts captured CO₂ into CO, which is then mixed with H₂ to reach H₂/CO ≈ 2 .

A6. A fuel cell with a molten lithium/potassium carbonate electrolyte operates at about:

Answer C, 650 °C.

Justification. The MCFC electrolyte must be molten to conduct CO_3^{2-} ions, which requires about 600–700 °C.

A7. Overall power-to-methanol/DME efficiency given in the course:

Answer B, 50–55 %.

Justification. Electrolysis (60–70 % LHV) is followed by synthesis and distillation losses. Check: 0.189 t H₂ per tonne of methanol at about 55 kWh/kg gives 10.4 MWh, against 5.5 MWh of methanol (LHV 19.9 GJ/t), that is about 53 %.

A8. For a plant driven mainly by a solar park, like Kassø, the best-suited electrolyser is:

Answer D, PEM, because of its response in seconds and wide partial-load range.

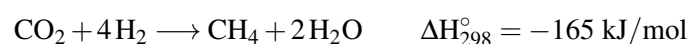
Justification. Solar output varies within minutes and falls to zero at night. PEM follows these variations; AWE has a minimum load of 15–40 % and slower dynamics, and SOEC needs a steady high-grade heat source.

Part B. True or False**(8 × 0.5 = 4 pts)**

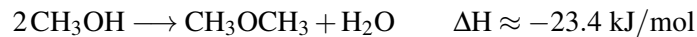
No.	Statement	T/F	Correction or justification
B1	Electrolytic hydrogen routinely reaches 99.99 % purity.	T	Electrolysis gives very pure H ₂ (99.9 to 99.999 % after drying and deoxidation), suitable for fuel cells.
B2	Direct air capture extracts CO ₂ from ambient air at about 420 ppm.	T	The very low concentration is why DAC is the most energy-intensive and costly CO ₂ source.
B3	Alkaline electrolyzers need platinum-group metal catalysts.	F	Correction: AWE uses non-noble nickel-based electrodes; PEM needs platinum (cathode) and iridium (anode).
B4	Methanation CO ₂ + 4H ₂ → CH ₄ + 2H ₂ O is endothermic.	F	Correction: it is strongly exothermic, ΔH ₂₉₈ ^o = −165 kJ/mol; heat must be removed.
B5	In an alkaline fuel cell (AFC), water is produced at the cathode.	F	Correction: water is produced at the anode: H ₂ + 2OH [−] → 2H ₂ O + 2e [−] .
B6	Biogenic CO ₂ supply may be more dispersed and seasonal than industrial point sources.	T	Biogas and bioethanol plants are small, scattered and depend on feedstock seasons, unlike cement or steel plants.
B7	The round-trip efficiency of Power-to-Power (30–45 %) is higher than that of batteries.	F	Correction: it is lower; batteries reach about 80–95 %. Power-to-Power is justified for long-duration and seasonal storage.
B8	Without heat recovery, power-to-SNG efficiency is 50–65 %; with heat recovery it can exceed 70–80 %.	T	Recovering the methanation and electrolyser heat (district heating, process heat) raises the useful-energy efficiency.

Part C. Complete the missing information**(3 pts)****Electrolyser technology comparison matrix (answers in green).**

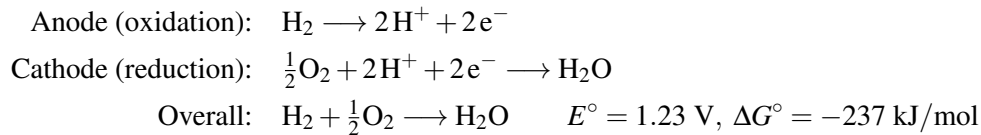
Factor	Alkaline (ALK)	PEM	AEM	SOEC
Electrolyte	Aqueous KOH (25–30 %)	Solid polymer	Solid anion membrane	Ceramic YSZ (O^{2−}-conductor)
Operating temperature	60–90 °C	50–80 °C	40–70 °C	600–1000 °C
Response time	Slow	Fast	Fast	Slow
Maturity	Mature (commercial for decades)	Commercialized	Emerging	Pre-commercial
Lifetime (h)	60,000–90,000	30,000–90,000	10,000–20,000	< 20,000 (about 10,000–20,000)

Part D. Reactions**(3 pts)****D1. Complete and name CO₂ + 4H₂ → ...****(0.75 pt)**

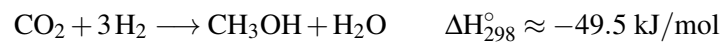
Sabatier reaction (catalytic CO₂ methanation), the core of Power-to-Methane. Nickel-based catalyst (ruthenium as an alternative), about 250–400 °C and 1–30 bar. The reaction is exothermic and reduces the number of gas moles (5 to 3), so it is favoured by moderate temperature and higher pressure.

D2. Methanol to dimethyl ether (DME) and type of reaction (0.75 pt)

Catalytic dehydration (condensation) of methanol over a solid acid catalyst ($\gamma\text{-Al}_2\text{O}_3$ or zeolite), about 250–400 °C and 10–20 bar. It is mildly exothermic.

D3. Half-reactions and overall reaction of a PEM fuel cell (0.75 pt)

Protons cross the polymer membrane; water forms at the cathode; operation at about 60–80 °C.

D4. Kassø synthesis reaction and reactant masses per tonne of methanol (0.75 pt)

Cu/ZnO/Al₂O₃ catalyst, about 200–300 °C and 50–100 bar.

$$\begin{aligned} n(\text{CH}_3\text{OH}) &= \frac{10^6 \text{ g}}{32.04 \text{ g/mol}} = 31,211 \text{ mol} \\ m(\text{H}_2) &= 3 \times 31,211 \times 2.016 = 188,800 \text{ g} \approx \mathbf{0.189 \text{ t}} \\ m(\text{CO}_2) &= 31,211 \times 44.01 = 1,373,600 \text{ g} \approx \mathbf{1.374 \text{ t}} \end{aligned}$$

Each tonne of methanol therefore needs about 189 kg of H₂ and 1.37 t of CO₂, and releases 0.56 t of water. In practice the consumption is slightly higher because of purge losses and side products.

Part E. Case study: Project 4, Kassø e-Methanol (Denmark) (6 pts)**E1. Name, country, developers or owners and main objective (1 pt)**

- **Name and location:** Kassø e-methanol facility (Kassø Power-to-X), Kassø, near Aabenraa, Southern Jutland, Denmark.
- **Developer and owners:** developed by the Danish renewable-energy company European Energy, owned 51 % by European Energy and 49 % by Mitsui & Co. (Japan). Electrolysers supplied by Siemens Energy; methanol loop designed and built by European Energy.
- **Milestones:** first green hydrogen in January 2025, first e-methanol in March 2025, RFNBO certification (ISCC EU) in April 2025, official inauguration on 13 May 2025.
- **Main objective:** produce up to 42,000 t/yr of e-methanol at the world's first commercial-scale e-methanol plant, to decarbonise shipping (A.P. Moller-Maersk) and chemical and plastic production (LEGO Group, Novo Nordisk; OMV as additional offtaker), and to demonstrate Power-to-X at industrial scale.

E2. Power-to-X pathway as a chain of steps and core reaction (1.5 pts)

1. Renewable electricity: 304 MW Kassø solar park (about 50 %) and grid electricity under a balancing agreement (about 50 %).
2. Water treatment to deionised quality, then PEM electrolysis (three units, about 52 MW): $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$.
3. Hydrogen compression and buffer storage, which decouples variable electrolysis from the synthesis loop.
4. Biogenic CO₂ captured at biogas plants (the first batch came from Tønder), purified and delivered.
5. Catalytic methanol synthesis in a recycle loop, then distillation to remove water.
6. Storage and delivery of e-methanol to offtakers.

Core reaction: $\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$ ($\Delta H \approx -49.5 \text{ kJ/mol}$). Combined with electrolysis, the net Power-to-Methanol balance is $\text{CO}_2 + 2\text{H}_2\text{O} \rightarrow \text{CH}_3\text{OH} + \frac{3}{2}\text{O}_2$.

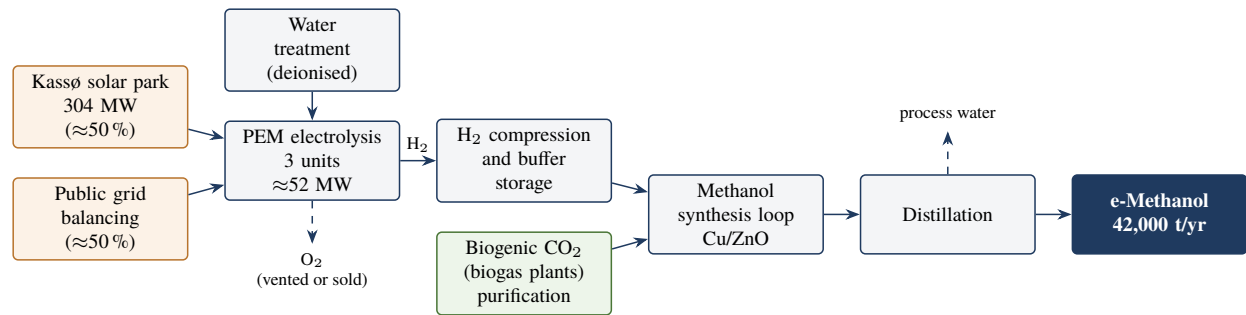


Figure 1: Simplified process chain of the Kassø e-methanol plant (dashed arrows: by-products).

E3. Key technical data and choice of electrolyser

(1.5 pts)

Item	Value
Capacity	42,000 t/yr e-methanol (about 53 million litres)
Electrolysis	PEM, Siemens Energy, three units, about 52 MW in total
Hydrogen	about 6,000 t/yr (project page) to 8,000 t/yr (2025 media kit); stoichiometry for 42,000 t requires 7,930 t
Power supply	304 MW solar park (about 50 %) and grid (about 50 %)
Carbon source	Biogenic CO ₂ from biogas plants, about 57,700 t/yr at nameplate
Product quality	Industry-grade e-methanol, certified RFNBO under the EU scheme ISCC EU

Why PEM. The plant is driven by solar power, which varies within minutes and stops at night. PEM electrolyzers respond in seconds, operate over a wide partial-load range, start quickly, deliver high-purity hydrogen at elevated pressure and are compact. Alkaline electrolyzers are cheaper but have a minimum load of 15–40 % and slower dynamics; SOEC requires a steady high-temperature heat source that a solar-driven site does not have. The cost of PEM (platinum and iridium catalysts, ultrapure water) is accepted in exchange for flexibility.

E4. Improvement question: which inputs would you change?

(2 pts)

The plant has four inputs: electricity, water, CO₂ and heat. In order of impact:

- Electricity.** Hybridise the supply with dedicated wind power purchase agreements. Danish solar has a capacity factor of only about 11–12 %, while wind reaches 30–45 %; a hybrid profile raises electrolyser full-load hours, lowers the cost per kilogram of hydrogen and secures hourly renewable matching, which EU RFNBO rules require from 2030.
- Heat.** Recover electrolyser waste heat (30–40 % of the input, at 50–80 °C) and the exothermic synthesis heat for distillation and district heating. This raises the useful-energy efficiency well above the 50–55 % of the methanol product alone.
- Water.** Recycle the 0.56 t of process water released per tonne of methanol to the electrolyzers. At nameplate this is about 23,600 t/yr, roughly one third of the theoretical demand of about 71,000 t/yr.
- CO₂.** Contract several biogenic sources and add liquid CO₂ storage, since biogenic supply is dispersed and seasonal; direct air capture can be a long-term complement.
- By-product oxygen.** Sell the oxygen (about 8 kg per kg H₂, around 60,000 t/yr) to wastewater, medical or industrial users to add revenue.

End of Subject A

Model Answers | Subject B

Power-to-X (PtX)

Part A. Multiple-choice questions

(8 × 0.5 = 4 pts)

A1. The first documented Power-to-X projects appeared around:

Answer B, 1990.

Justification. Reviews of demonstration projects list the first plants from the late 1980s and early 1990s (for example Higashi-Hiroshima 1988, Fayetteville 1991, PHOEBUS in Jülich 1993). 1839 is Grove's fuel cell, not a PtX project.

A2. In an AWE, the electrolyte is typically:

Answer A, 25–30 % aqueous KOH.

Justification. Alkaline electrolyzers use a concentrated potassium hydroxide solution and nickel electrodes; OH^- is the charge carrier.

A3. Theoretical water consumption of electrolysis:

Answer B, 9 kg H_2O per kg H_2 .

Justification. From $\text{H}_2\text{O} \rightarrow \text{H}_2 + \frac{1}{2}\text{O}_2$: $18.015/2.016 = 8.94$ kg of water per kg of hydrogen. Real plants need about 10–15 kg because of water treatment losses.

A4. Typical operating conditions of Haber–Bosch synthesis:

Answer D, 150–300 bar, 400–500 °C, iron-based catalyst.

Justification. High pressure shifts the equilibrium towards ammonia (4 moles of gas give 2); 400–500 °C gives acceptable kinetics on the promoted iron catalyst. Option B describes methanol synthesis.

A5. CO_2 source with a high concentration (10–30 % or more):

Answer B, industrial point sources (cement, steel, refineries).

Justification. The higher the CO_2 concentration, the lower the capture energy and cost. Air contains only about 420 ppm.

A6. To stay competitive without heavy subsidies, renewable electricity should ideally cost less than:

Answer B, 30–40 €/MWh.

Justification. Electricity is the largest cost item of green hydrogen: at about 55 kWh/kg, each 10 €/MWh adds about 0.55 €/kg H_2 .

A7. Why is intermediate H_2 storage recommended between electrolyzers and synthesis units?

Answer A, to buffer slower synthesis units while fast electrolyzers track variable renewable surplus.

Justification. Synthesis loops (methanol, ammonia, methanation) prefer steady operation and ramp slowly, whereas the electrolyser follows the renewable profile; the buffer decouples the two.

A8. Share of global anthropogenic CO_2 from conventional ammonia production:

Answer B, 1.2 %.

Justification. Conventional ammonia relies on steam methane reforming or coal gasification and accounts for roughly 1–2 % of global CO_2 emissions.

Part B. True or False**(8 × 0.5 = 4 pts)**

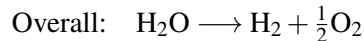
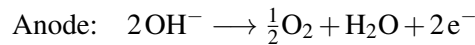
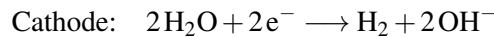
No.	Statement	T/F	Correction or justification
B1	More than 80 % of current final energy still comes from fossil fuels.	T	Fossil fuels still supply about four fifths of global energy, which motivates Power-to-X for hard-to-electrify uses.
B2	Power-to-Heat is a hydrogen-based PtX route treated in detail in the series.	F	Correction: Power-to-Heat converts electricity directly into heat without hydrogen; the series focuses on hydrogen-based routes.
B3	AWE has a faster dynamic response than PEM.	F	Correction: PEM is faster (seconds); AWE is slower and has a minimum load of 15–40 %.
B4	Ammonia combustion or cracking produces no CO ₂ , only N ₂ and water (NO _x must be controlled).	T	NH ₃ contains no carbon: $4\text{NH}_3 + 3\text{O}_2 \rightarrow 2\text{N}_2 + 6\text{H}_2\text{O}$.
B5	In a PAFC, water is produced at the cathode and the cell tolerates CO ₂ in air.	T	The acid electrolyte conducts H ⁺ to the cathode, and unlike the alkaline AFC it does not react with CO ₂ .
B6	Biological methanation needs 200–400 °C and 30 bar.	F	Correction: biological methanation uses archaea at about 30–70 °C and near-ambient to moderate pressure (1–10 bar); 200–400 °C applies to catalytic methanation.
B7	Additionality means the renewable generation used must be new, not diverted from other consumers.	T	This is the EU RFNBO additionality principle (Delegated Regulation 2023/1184).
B8	Electrolyser flexibility always increases utilisation and therefore always lowers costs.	F	Correction: following variable power often lowers full-load hours and raises capital cost per kg; flexibility lowers cost only when the savings on electricity exceed this effect.

Part C. Complete the missing information**(3 pts)****Fuel-cell types** (answers in green).

Type	Electrolyte	Mobile ion	Temperature	Water produced at
AFC	Aqueous KOH	OH⁻	~60–90 °C	anode
PAFC	Liquid phosphoric acid (H₃PO₄)	H ⁺	~150–200 °C	cathode
PEMFC	Solid polymer membrane	H ⁺	~60–80 °C	cathode
MCFC	Molten Li/K carbonate	CO₃²⁻	~650 °C	anode
SOFC	Solid ceramic (zirconia)	O ²⁻	~800–1000 °C	anode

Part D. Reactions**(3 pts)****D1. Overall water electrolysis and gas formed at the cathode****(0.75 pt)**

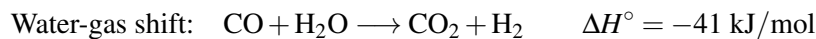
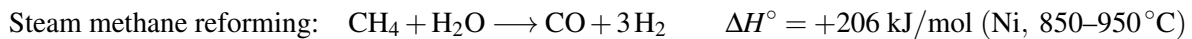
Reversible voltage 1.23 V, thermoneutral voltage 1.48 V. **Hydrogen forms at the cathode** (reduction) and oxygen at the anode, in a 2:1 volume ratio.

D2. Half-reactions of alkaline water electrolysis (0.75 pt)

OH^- ions migrate through the diaphragm from the cathode to the anode, in 25–30 % KOH at 60–90 °C.

D3. Haber–Bosch reaction, conditions and catalyst (0.75 pt)

Promoted iron catalyst (magnetite with K_2O and Al_2O_3), 150–300 bar, 400–500 °C. Conversion per pass is only about 15–25 %, so unreacted gas is recycled after ammonia condensation.

D4. Steam reforming and water-gas shift, and why they are not used at Kassø (0.75 pt)

These reactions make hydrogen and syngas from fossil methane and release about 9–10 kg of CO_2 per kg of H_2 . At Kassø, hydrogen comes from water electrolysis powered by renewable electricity, and carbon comes from biogenic CO_2 that is hydrogenated directly ($\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$). No fossil feedstock and no reforming step are needed, which is the condition for RFNBO status.

Part E. Case study: Project 4, Kassø e-Methanol (Denmark) (6 pts)**E1. Renewable energy sources and their proportion (1 pt)**

About **50 %** of the electricity comes from the adjacent **304 MW Kassø solar park**, one of the largest in Northern Europe. The other **50 %** comes from the **public grid** through a balancing agreement (Danske Commodities); in Denmark this supply is dominated by wind power. The grid share compensates for the low capacity factor of Danish solar (about 11–12 %) and keeps the electrolyzers running outside sunny hours.

E2. Environmental benefits and EU condition (1 pt)

Benefits. e-Methanol replaces fossil methanol made from natural gas or coal; it reuses biogenic carbon (short carbon cycle), uses surplus renewable power, and burns without SO_x and with low particulate and NO_x emissions, which suits shipping (Maersk) and plastics (LEGO, Novo Nordisk).

EU condition. The product must qualify as a **renewable fuel of non-biological origin (RFNBO)** under the Renewable Energy Directive:

- at least **70 % greenhouse-gas savings** against the fossil comparator of 94 g CO_2 -eq/MJ, that is at most 28.2 g CO_2 -eq/MJ (Delegated Regulation (EU) 2023/1185);
- renewable electricity rules (Delegated Regulation (EU) 2023/1184): additionality, temporal correlation (monthly until 2029, hourly from 2030) and geographical correlation (same bidding zone);
- certification by a recognised scheme. Kassø obtained ISCC EU RFNBO certification in April 2025, the first such certification for e-methanol.

E3. Mass balance (2 pts)

Reaction $\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$, with 1 mol CH_3OH (32.04 g) requiring 3 mol H_2 (6.048 g) and 1 mol CO_2 (44.01 g).

$$(a) \quad m(\text{H}_2) = 42,000 \times \frac{6.048}{32.04} = \mathbf{7,928 \text{ t H}_2/\text{yr}}$$

$$(b) \quad m(\text{CH}_3\text{OH}) = 6,000 \times \frac{32.04}{6.048} = \mathbf{31,786 \text{ t CH}_3\text{OH}/\text{yr}}$$

$$(c) \quad m(\text{CO}_2) = 42,000 \times \frac{44.01}{32.04} = \mathbf{57,691 \text{ t CO}_2/\text{yr}}$$

Comment.

- 6,000 t of H₂ can only give about 31,800 t of methanol, that is **76 %** of the nameplate. The 42,000 t/yr is a maximum capacity; reaching it needs about 7,930 t of H₂, which matches the 8,000 t/yr quoted in the 2025 media kit.
- With 52 MW at about 55 kWh/kg, full-load operation would give about 8,300 t H₂/yr. Producing 6,000 t corresponds to about 72 % of the hours, and 7,930 t to about 96 %: solar alone cannot achieve this, hence the grid supply.
- The plant consumes about 1.37 t of biogenic CO₂ per tonne of methanol, and the same carbon is released when the fuel is burnt: the benefit comes from replacing fossil carbon, not from permanent storage.
- Real consumption is slightly higher than stoichiometry because of purge gas losses and by-products.

E4. Improvement question: which inputs would you change? (2 pts)

1. **Electricity mix.** Add wind PPAs and possibly a battery to raise electrolyser full-load hours (solar capacity factor of about 11–12 % against 30–45 % for wind) and to meet the hourly matching rule from 2030. This acts on the largest cost item, electricity.
2. **Hydrogen buffer.** Enlarge hydrogen storage so the synthesis loop runs steadily while the electrolyzers follow the renewable profile.
3. **Heat and water integration.** Use electrolyser and synthesis heat for distillation and district heating, and recycle the 0.56 t of process water per tonne of methanol to the electrolyzers.
4. **CO₂ supply.** Diversify biogenic sources and add liquid CO₂ storage to cover seasonal variation.

End of Subject B

Model Answers | Subject C

Power-to-X (PtX)

Part A. Multiple-choice questions

(8 × 0.5 = 4 pts)

A1. Fuel cell used on the Apollo spacecraft:

Answer D, alkaline fuel cell (AFC).

Justification. Apollo used pure H₂ and O₂, so the CO₂ sensitivity of the KOH electrolyte was not a problem; the cells also supplied drinking water.

A2. The fuel cell was demonstrated in 1839 by:

Answer C, William Robert Grove.

Justification. Grove built the “gas voltaic battery”. Schönbein described the effect in 1838; Haber and Sabatier are linked to ammonia and methanation.

A3. Seasonal mismatch, which batteries cannot economically solve, is typical of:

Answer B, mid- and high-latitude climates where winter demand peaks when solar is lowest.

Justification. Batteries store hours to days; shifting energy over months requires chemical carriers such as hydrogen or methane.

A4. The Fischer–Tropsch synthesis uses catalysts based on:

Answer A, Co or Fe.

Justification. Cobalt (low-temperature FT, long-chain waxes and diesel) and iron (high-temperature FT, tolerant to CO₂). Cu/ZnO is for methanol, Ni for methanation, Pt for PEM cells.

A5. Dry reforming of methane is:

Answer D, $\text{CH}_4 + \text{CO}_2 \rightarrow 2\text{CO} + 2\text{H}_2$.

Justification. “Dry” means CO₂ replaces steam; the reaction is strongly endothermic ($\Delta H \approx +247$ kJ/mol) and gives H₂/CO = 1.

A6. In a SOEC, the species that crosses the electrolyte is:

Answer B, O^{2−}.

Justification. Oxide ions formed at the cathode ($\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + \text{O}^{2-}$) cross the YSZ ceramic to the anode.

A7. Most HRES studies still treat PtX as:

Answer A, a simple efficiency black-box.

Justification. Many hybrid renewable energy system models use a fixed conversion efficiency and ignore part-load behaviour, start-up times, degradation and coupling between energy carriers.

A8. Correct statement about Power-to-Power:

Answer A, reversible solid oxide cells can do both electrolysis and fuel-cell operation.

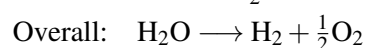
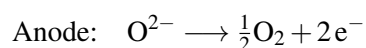
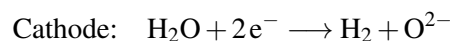
Justification. Its round-trip efficiency is only about 30–45 %, it targets long-duration storage, and it does not require pumped hydro.

Part B. True or False**(8 × 0.5 = 4 pts)**

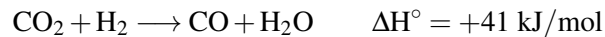
No.	Statement	T/F	Correction or justification
B1	Typical wind and solar capacity factors are 20–45 %.	T	Onshore wind about 25–45 %, solar about 15–25 % in sunny regions (lower at high latitude, about 11–12 % in Denmark).
B2	Ammonia contains carbon.	F	Correction: NH ₃ contains only nitrogen and hydrogen; it is a carbon-free carrier.
B3	PEM electrolyzers need ultra-pure feed water.	T	Deionised water (conductivity below about 1 μS/cm) protects the membrane and catalysts.
B4	SOEC has the fastest start-up of all electrolyzers.	F	Correction: SOEC is the slowest (hours of heating to 600–1000 °C); PEM starts fastest.
B5	FT-SAF is chemically similar to conventional jet fuel and can be used in existing aircraft engines with minimal change.	T	FT synthetic paraffinic kerosene is a drop-in fuel, approved for blends up to 50 % (ASTM D7566).
B6	In the AEM electrolyser, hydroxide ions migrate to the anode.	T	OH [−] formed at the cathode crosses the anion-exchange membrane and is oxidised to O ₂ at the anode.
B7	PtL overall efficiencies of 40–55 % are higher than electrolysis efficiencies of 60–70 % (LHV).	F	Correction: they are lower, because each synthesis step adds losses after electrolysis.
B8	Power-to-X can turn curtailed renewable electricity into new revenue streams and controllable demand.	T	Electrolyzers absorb surplus power that would otherwise be curtailed and provide flexibility to the grid.

Part C. Complete the missing information**(3 pts)****Key reactions for syngas production** (answers in green).

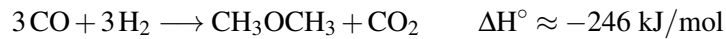
Process	Reaction	Typical temperature
Steam reforming	$\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$	850–950 °C
Dry reforming	$\text{CH}_4 + \text{CO}_2 \rightarrow 2\text{CO} + 2\text{H}_2$	about 800–1000 °C
Partial oxidation	$\text{CH}_4 + \frac{1}{2}\text{O}_2 \rightarrow \text{CO} + 2\text{H}_2$	catalytic 800–900 °C
Water-gas shift	$\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$	350–450 °C (Fe–Cr) / 200–250 °C (Cu–Zn)
Coal / biomass gasification	$\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$	about 800–1000 °C (up to 1400 °C in entrained-flow gasifiers)

Part D. Reactions**(3 pts)****D1. Half-reactions of a SOEC****(0.75 pt)**

Steam is fed at 600–1000 °C; part of the energy is supplied as heat, which lowers the electricity demand (about 40 kWh/kg H₂).

D2. Reverse water-gas shift and its role in Power-to-Liquid**(0.75 pt)**

Endothermic, favoured above about 700 °C. It converts captured CO₂ into CO, so that green H₂ and CO can be mixed into syngas with H₂/CO ≈ 2 for Fischer–Tropsch synthesis of e-kerosene and e-diesel.

D3. Direct DME synthesis from syngas**(0.75 pt)**

One-step process on a bifunctional catalyst (Cu/ZnO for methanol synthesis combined with a solid acid for dehydration), about 240–280 °C and 30–50 bar. The water-gas shift consumes the water produced, which pushes conversion further.

D4. Methanation versus methanol synthesis**(0.75 pt)**

	Methanation	Methanol synthesis
Reaction	$\text{CO}_2 + 4 \text{ H}_2 \rightarrow \text{CH}_4 + 2 \text{ H}_2\text{O}$	$\text{CO}_2 + 3 \text{ H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$
Enthalpy	–165 kJ/mol	–49.5 kJ/mol
Catalyst	Ni (Ru)	Cu/ZnO/Al ₂ O ₃
Conditions	250–400 °C, 1–30 bar	200–300 °C, 50–100 bar
H ₂ per mole CO ₂	4 mol	3 mol
Product	Gas (SNG), injectable in gas grids	Liquid, easy to store and ship

Methanol synthesis uses 25 % less hydrogen per mole of CO₂ and gives a liquid product, while methanation is more exothermic and needs efficient heat removal.

Part E. Case study: Project 4, Kassø e-Methanol (Denmark)**(6 pts)****E1. Main technologies (unit operations) in process order****(1 pt)**

1. Renewable power supply: 304 MW solar park and grid connection, transformers and rectifiers (AC to DC).
2. Water treatment: filtration and demineralisation to PEM-grade water.
3. PEM electrolysis: three Siemens Energy units, about 52 MW in total.
4. Gas conditioning: hydrogen drying, compression and buffer storage; oxygen vented or valorised.
5. CO₂ supply chain: capture at biogas plants, purification, liquefaction, transport and storage.
6. Feed compression and mixing to the stoichiometric ratio.
7. Catalytic methanol synthesis loop (Cu/ZnO) with recycle of unconverted gas and purge.
8. Flash separation and distillation to remove water and by-products.
9. Product storage and dispatch to offtakers.

E2. Three economic or technical challenges**(1 pt)**

1. **Cost competitiveness.** Renewable e-methanol cost about USD 800–1,600/t against USD 100–250/t of production cost for fossil methanol (IRENA and Methanol Institute, 2021). Electricity and electrolyzers dominate this gap, so long-term offtake contracts (Maersk, LEGO, Novo Nordisk) are essential.
2. **Variable supply against a steady synthesis loop.** Solar has a capacity factor of only about 11–12 % in Denmark, while the methanol loop needs stable operation; this requires grid balancing power, hydrogen buffer storage and careful control.
3. **CO₂ logistics and regulation.** Biogenic CO₂ is dispersed and seasonal, and the plant must continuously prove RFNBO compliance (70 % GHG savings, additionality, temporal and geographical correlation, with hourly matching from 2030).

E3. Comparison with conventional methanol production**(2 pts)**

Criterion	Conventional methanol	Kassø e-methanol
Energy source	Natural gas (feed and fuel, about 30 GJ per tonne) or coal	Renewable electricity, about 10.4 MWh per tonne for the hydrogen (0.189 t H ₂ at about 55 kWh/kg)
Hydrogen route	Steam or autothermal reforming of methane, coal gasification	Water electrolysis (PEM)
Carbon source	Fossil carbon in natural gas or coal	Biogenic CO ₂ from biogas plants
Emissions	Fossil comparator 94 g CO ₂ -eq/MJ, that is about 1.9 t CO ₂ -eq per tonne of methanol over the life cycle	At most 28.2 g CO ₂ -eq/MJ (at least 70 % savings, RFNBO certified)
Scale	World-scale plants of about 5,000 t/day (about 1.7 Mt/yr); global output about 100 Mt/yr	42,000 t/yr (about 115 t/day), roughly 40 times smaller than a world-scale plant
Cost	About USD 100–250/t (production cost)	About USD 800–1,600/t (IRENA, 2021)

The chemistry downstream of the synthesis gas is the same (Cu/ZnO catalyst, 50–100 bar, distillation); the difference lies in where the hydrogen and the carbon come from. e-Methanol is cleaner but several times more expensive, which is why regulation (RFNBO, FuelEU Maritime) and long-term offtake contracts drive its market today.

E4. Improvement question: prioritised list of input changes**(2 pts)**

- Electricity, priority 1.** Combine the solar park with wind PPAs (and a battery if needed) to raise electrolyser full-load hours, reduce the cost per kilogram of H₂ and secure hourly renewable matching from 2030. Electricity is the largest cost item.
- Hydrogen buffer, priority 2.** Increase H₂ storage between the electrolyzers and the synthesis loop, so that the loop runs at steady load while the electrolyzers follow renewable output.
- Heat, priority 3.** Recover electrolyser heat (30–40 % of the input) and the synthesis heat (–49.5 kJ/mol) for distillation and district heating, raising the overall useful-energy efficiency.
- Water, priority 4.** Recycle the process water (0.56 t per tonne of methanol, about one third of the theoretical electrolysis demand) and recover water from the distillation column.
- CO₂, priority 5.** Diversify biogenic sources and add buffer storage of liquid CO₂; consider direct air capture in the long term.
- Oxygen, priority 6.** Sell by-product oxygen (about 8 kg per kg H₂) to industrial or medical users.

End of Subject C

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